AGRICULTURAL AI INTERGRATED SYSTEM(Agrilink ai)

SUBMITTED BY,

Jim Carson Wagune

DSE-03-8769/2024

A RESEARCH SYSTEM PROPOSAL SUBMITTED IN FULFILMENT FOR PROGRRESSION TO SYSTEM PROJECT TO THE DEPARTMENT OF ICT AND ENGENEERING OF ZETECH UNIVERSITY

November,2025

DECLARATION

I, Jim Carson Wagune ,DSE-03-8769/2024,hereby declare that this research proposal entitled "Agricultural AI Integrated System (Agrilink AI)" is my original work and has not been submitted by any student or researcher for academic credit .Any material or quoted from other sources have been acknowledged through proper citation and referencing

Student Name:Jim Carson Wagune

Signature:

ID Number:DSE-03-8769/2024

Date:

DEDICATION

This work I whole hearted dedicate it to my loving family, whose unweaving encouragement has been my foundation throughout my academic journey. To all hardworking farmers across the globe who continue to feed the world with their crops and products, this project is a tribute to your resilience and innovation. May Agrilink AI contribute your efforts easier, smarter and more productive

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of this project proposal. Special thanks to my supervisor, MR. Barrack Ouko, for his priceless encouragement and constructive feedback. I am also thankful to my classmates and friends, whose dependable support made this work possible. Lastly, I'm thankful to the institutions and reference materials that contributed meaningfully to the conceptual and technical grounding of this document

ABSRACT

Agricultural Ai integrated system(Agrilink Ai) is a technology that will bring a rapid change to the agricultural environment .The system enables an increases in agricultural production and dependency since the system will be providing the farmers with all they need all in one.Agrilink Ai system improve market accessibility, enhance crop productivity and strengthen farmers network through collaborations dashboards. The system ability to diagnose diseases and help farmer

TABLE OF CONENTS

Contents

DECLARATION ii

DEDICATION iii

ACKNOWLEDGEMENT iv

ABSRACT v

TABLE OF CONENTS vi

LIST OF FIGURES ix

LIST OF TABLES x

ABREVIATIONS AND ACRONMYS xi

1.1 Introduction 1

1.2 Background of the study 1

1.3 Problem Statement 2

1.4 Purpose of the study 2

1.5 Objectives 3

1.5.1 General Objectives 3

1.5.2 Specific Objectives 3

1.6 Research Questions 3

1.7 Significance of the study 3

1.8 Scope of the study 4

1.8.1 The study will concentrate on the following key areas: 4

2.1 Introduction 5

2.2 Overview of Artificial Intelligence in Agriculture 5

2.3 AI-Based Pest and Disease Diagnosis 5

2.4 AI in Market Linkage and Supply Chain Optimization 6

2.5 Decision Support Systems(DSS)in Agriculture 6

2.6 Farmer Collaboration and Knowledge-Sharing Platforms 6

2.7 Existing Agricultural AI Systems 7

2.8 Research Gaps 7

2.9 Summary 7
References 8
3.1 Introduction 9
3.2 Research/Development design 9
3.2.1 Requirements Analysis 9
3.2.2 System Design 9
3.2.3 Implementation (Coding) 10
3.2.4 Testing 10
3.2.5 Deployment 10
3.2.6 Maintenance 10
3.3 Tools and technologies used 10
3.3.1 Programming languages 10
3.3.2 AI Learning Libraries 10
3.3.3 Database System 11
3.3.4 APIs and Framework 11
3.3.5 Frontend Tolls 11
3.4 System Architecture 11
3.4.1 Overall System Architecture 11
i. Presentation Layer 11
ii. Application Layer 11
iii. Database Layer 12
3.5 Conceptual System Architecture Diagram 12
3.5.1 Use Case Diagram 13
3.5.2 Data Flow Diagram (DFD)Level 0 14
3.5.3 Data Flow Diagram (DFD) Level 1 (AI Diagnosis Module) 15
1.5.4 Entity Relationship Diagram (ERD) 16
3.5.5 Logical Design Diagram 17
3.5.6 Physical Design Diagram (Network Architecture) 18
3.5 Data Collection Methods 18

3.5.1 Primary Data Collection 19
i. Interviews 19
ii. Questionnaires 19
iii. Observation 19
3.5.2 Secondary Data Collection 19
3.6 Data Analysis 19
3.7 Ethical Consideration in Data Collection 20
3.8 System Implementation 20
3.9 Testing and Evaluation 20
3.10 Ethical Considerations 20
3.11 Summary 20
References 21
Appendices 22

LIST OF FIGURES

Figure 1 13
Figure 2 14
Figure 3 15
Figure 4 16

LIST OF TABLES

Table 1 12
Table 2 12
Table 3 12
Table 4 17
Table 5 17

ABREVIATIONS AND ACRONMYS

AI-Artificial Intelligence

APIs- Application Programming Interfaces

Agrilink: Agricultural AI integrated system

CHAPTER ONE:INTRODUCTION

1.1 Introduction

Agriculture, or the study and practice of cultivating food, has traditionally been said to be the foundation of most economies of the plummeting world. With the possibility of meeting basic needs and food security for the population, the income depends on agriculture in these countries. Yet, are increasingly acknowledged as impediments outside adequately remunerating, productive, and cheap? Some of these problems are dependency on middlemen for access to markets, pest and weather outbreaks, obsolete agricultural-information technology, and the need for enhanced social collaboration between farmers.

Artificial Intelligence (AI), once considered a distant notion or a far-off possibility, now opens a window to smart applications that can revolutionize agriculture I'm positive. Systems powered by AI could diagnose crop diseases, provide decision-making support, and prepare digital platforms that connect farmers with customers so that worthless middlemen could be ideally eliminated. AI, by ensuring co-creative dashboards, may also empower farmers in sharing knowledge and solving problems collectively

1.2 Background of the study

Traditionally, farmers have relied on middlemen to access markets, which often results in exploitation and reduced profit margins. Moreover, frequent outbreaks of pests and crop diseases cause significant yield losses, threatening both food security and farmers' livelihoods. Limited access to agricultural extension services and decision-support tools further compounds these problems, leaving farmers unable to make informed decisions regarding planting, harvesting, and marketing. In addition, weak collaboration among farmers reduces opportunities for knowledge-sharing, innovation, and collective problem-solving.

Agrilink AI is proposed as an integrated AI-based agricultural system designed to improve market accessibility, enhance crop productivity, and strengthen farmer networks. The system will connect farmers directly with customers, provide AI-powered pest and disease diagnosis, support better decision-making, and create organizational dashboards to build collective knowledge. By addressing existing barriers in agriculture, Agrilink AI aims to promote efficiency, increase profitability, and contribute to sustainable agricultural development.

1.3 Problem Statement

To give those engaging in farming as living, and now struggles with problems in their agricultural production . Markets become problematic for farmers as middlemen stand in the way of profit maximization. Crop productivity and food security are threatened by pest and disease outbreaks. In addition, the information that the farmer received is untimely and unreliable, so that the farmer progresses poorly in making decisions through production and marketing efficiencies. But poor connectivity and accessibility limit farmers in sharing knowledge, collaborating, and using digital solutions. These challenges imply the design of an agricultural AI integrated system for disease diagnosis, market linkage, decision support, and collaboration platforms to farmers.

1.4 Purpose of the study

The purpose of this study is dedicated toward designing and implementing an integrated AI-enabled agriculture system which we have named Agrilink AI that has the potential of mitigating some of these longstanding challenges faced by farmers in the agriculture sector especially those in developing regions. Agriculture still occupies the bulk of rural people livelihoods though productivity and profitability are characterized by labor exploitative market, pests and diseases occurrence, lack of ready access to accurate low cost agricultural information especially in remote areas as well as weak knowledge sharing networks among famers.

The study aims in exploring the role of Artificial Intelligence technologies in revolutionizing farming, for the purpose of automating disease diagnosis and increasing agriculture productivity among others. Leveraging AI tools, Agrilink AI is a one-stop digital platform where farmers can directly communicate with consumers, instantly request pest and disease diagnostic symptoms, make use of smart farming insights, and share knowledge on how to grow crops fast by connecting through interactive dashboards.

The purpose of development of this system is to:

- Establish direct market links to promote fair trade and increase farmers' incomes to limit farmers' reliance on middlemen.
- Use AI-enabled early detection systems to maximize losses by timely and early interventions to increase crop productivity.
- Predict and analyze data against key variables to provide decision-making support in planting, harvesting, and marketing.

- Share digital dashboards designed to promote communication, idea circulation, and collaborative capacity building.

Transformative use of AI in re-engineering agricultural systems is the aim of this study. Outcomes will focus on sustainable agricultural development, food security, and farmers' economic empowerment. The study proposes a technological blueprint to policymakers, agricultural institutions, and innovators to develop context-specific AI agricultural interventions.

1.5 Objectives

1.5.1 General Objectives

i. To design an agricultural AI integrated system for farmers that facilitate efficiency in production, decision making, Pest and disease diagnosis and knowledge-sharing among farmers

1.5.2 Specific Objectives

I. Improve Market by connecting farmers directly to the customers
II. Enhance agricultural production by ai diseases diagnosis
III. To improve farmers decision making
IV. Join farmers together through organization dashboards for build up knowledge-sharing

1.6 Research Questions

I. How can AI integrated system reduce the farmers dependency on middlemen to enhance direct access to the market and customers?
II. How can the system help in pest and diseases control and diagnose to improve production in agriculture ?
III. How can AI integrated system improve a farmers decision making and produce a good outcome?
IV. What is the role of organization dashboard in enhancing knowledge-sharing among farmers?

1.7 Significance of the study

This study will provide a farmer with direct market access ,reliability to agricultural knowledge ,early pest and diseases.Throught the integration of AI the farmer can make informed decisions .The finding will serve as a references for future researchers in he field of AI and agriculture innovation. Study bridges the gap between traditional and modern digital agricultural systems.

1.8 Scope of the study

The study mainly will concentrate how Artificial Intelligence (AI) can be applied to solve common problems in agricultural production such as :market middlemen, lack of information, Pest and disease .

1.8.1 The study will concentrate on the following key areas:

- AI-Powered pest and diseases diagnosis
- Market linkage and accessibility
- Decision support system
- Organizational dashboards for knowledge-sharing

CHAPTER TWO: LITURETURE REVIEW

2.1 Introduction

This chapter describes relevant literature and previous studies that concern the use of Artificial Intelligence (AI) in agriculture. It specifically examines how AI technologies have been employed for the diagnosis of pests and diseases, market linkage, decision support, and platforms for sharing information among farmers as well as identifying gaps in the current agricultural systems and justifying the need for the proposed Agrilink AI integrated system.

2.2 Overview of Artificial Intelligence in Agriculture

AI has gradually made inroads into agriculture through the automation of its decision-making processes and enhancing its productivity. The application of AI techniques in agriculture includes many such as predicting the disease of plants, yield forecasting, and resource optimization using machine learning, computer vision, and predictive analysis. (Liakos et al., 2018). According to the FAO (2022), AI-based applications are possible tools for creating a greener and more data-driven agricultural environment.

Taking a few concrete examples, AI models analyze images of leaves to check for the occurrence of diseases; predictive algorithms use data on weather and soil to drive instrumentation on irrigation and fertilization. It significantly reduces human errors and increases efficiency in farming operations.

2.3 AI-Based Pest and Disease Diagnosis

Diseases in crops are one of the main causes of low productivity in farming. Traditional detection of disease is by visual observation, which is labor-intensive and prone to mistakes. Computer vision and AI have in recent studies proven that it is possible to detect diseases automatically based on pictures of leaves from plants (Zhang et al., 2021).

Machine learning models trained using infected and normal plant datasets are capable of diagnosing diseases such as maize leaf blight, coffee rust, and tomato mosaic virus with accurate accuracy. TensorFlow and PyTorch are tools that allow developers to train convolutional neural networks (CNNs) that have the capability of diagnosing plant diseases in real time through smartphone cameras.

Moreover most existing AI systems are stand-alone and do not interact with farmer support tools or marketplaces—creating a gap that Agrilink AI is attempting to cover.

2.4 AI in Market Linkage and Supply Chain Optimization

Market linkage remains one of the biggest challenges facing farmers in developing nations. The majority of smallholder farmers, according to (FAO, 2022), depend on middlemen who fix prices and take away farmers' margins. AI-based market systems can make it easy to link farmers to consumers or buyers directly by predicting market demand, suggesting fair prices, and making transactions easier (Platform, 2023).

Some AI models have been developed to forecast commodity prices and match producers with buyers using real-time data analysis. Not many, though, have managed to combine AI market forecasts with integrated farmer–consumer platforms. Agrilink AI aims to fill this gap by providing direct digital connection between consumers and farmers in its integrated dashboard.

2.5 Decision Support Systems(DSS)in Agriculture

Decision Support Systems (DSS) enable farmers to base their agricultural choices on data when they decide about planting and irrigation and fertilization and harvesting practices. AI improves DSS operations through its ability to handle extensive data sets which originate from IoT sensors and satellite imagery and weather prediction systems (Documentation, 2024).

AI models through predictive analytics provide farmers with planting schedule recommendations and yield predictions and pest control advice. The research conducted by Liakos et al. (2018) discovered that AI-enabled DSS systems boost farm productivity by 25% during their pilot testing on farms. The progress of DSS tools continues to advance but smallholder farmers face two main obstacles because these systems remain out of their reach through expensive technology and inadequate digital skills. The Agrilink AI platform presents an affordable web-based solution which solves these two problems through an easy-to-use interface.

2.6 Farmer Collaboration and Knowledge-Sharing Platforms

The exchange of knowledge between farmers serves as an essential foundation for developing new agricultural methods. Through digital dashboards and forums farmers can share their experiences and solutions and effective techniques which help boost their farm productivity. Research conducted by Zhang (2021) shows that collaborative platforms help teams solve problems together while they implement new technologies.

The systems will improve through AI technology which will provide automatic discussion summaries and solution recommendations based on shared data and expert farmer connections. The organizational dashboard of Agrilink AI functions as a tool which enables farmers to work together through AI-generated data that supports their skill development and education.

2.7 Existing Agricultural AI Systems

Several existing AI systems have been developed globally to address agricultural challenges.

Plantix – a mobile app that uses AI to diagnose plant diseases from photos.

IBM Watson Decision Platform for Agriculture – provides weather forecasting and farm management insights.

Microsoft FarmBeats – integrates IoT and AI for precision agriculture.

FarmSmart – helps smallholder farmers plan and manage their crops through digital recommendations.

While the above arrangements have achieved extraordinary success, they frequently lack integration of disease diagnosis, trade access, and knowledge sharing, all of which are the main objectives of the Agrilink AI organization.

2.8 Research Gaps

From the literature reviewed, several gaps have been identified:

Atomization of existing solutions – new automated reasoning tools are prioritized over isolated problems such as disease detection or output prediction, although they need to be integrated with trade and social cooperation phases.

Accessibility problems – a number of machine intelligence agricultural solutions are planned for technologically advanced regions, leaving smallholder farmers in a developing area.

Restricted farmer participation – Few media facilitate corporate learning and interaction among farmers using artificial intelligence perceptions.

Agrilink ai addresses these intervals by developing a comprehensive, user-friendly, and accessible unmanned AI platform integrating market linkage, disease diagnosis, resolution support, and farmer exchange.

2.9 Summary

The current division is in charge of reviewing current writing on the use of AI in farming. The paper draws attention to important areas such as AI-powered disease diagnosis, retail links, choice support systems, and knowledge sharing stages. The rationale for the development of Agrilink's combined intelligence framework, which aims at uniting these functions in one accessible, user-friendly, and productive platform for farmers, is the holes found in the existing structures.

References

(Liako, 2018)
(Zhang Y. , 2021)
(Technology(FAO), 2022)
(Platform, 2023)
(Documentation T. , 2024)

CHAPTER THREE:METHODOLOGY

3.1 Introduction

This chapter will present the methodology that will be followed in the development of the Agrilink System online selling system. It will describe the research methodology, software development model, feasibility study, data collection techniques, tools and design diagrams. The purpose of this methodology is to provide a clear road map for successful system development and implementation.

3.2 Research/Development design

The Design & Development approach aims to design and evaluate a technological solution to handle real-world problems. The D&D approach was relevant to AgriLink AI, as it called for the design, development, and testing of an AI-based agricultural system in response to identified challenges in farming and market linkages.
I selected Waterfall method as the primary system development methodology this because , the process follows a linear and systematic method whereby the next phase could begin only after the preceding phase was completed. This makes sure that all stages of development, from analysis down to testing, are fully documented, controlled, and verified.
In this project, the stages of the Waterfall Model were:

3.2.1 Requirements Analysis

At this stage, the functional and non-functional requirements of the system were gathered by literature review, interviews with farmers, and consultations with agricultural experts. The requirements were documented with the intent to guide the architecture and design of the system. Some of the most important requirements included: Direct market linkage between farmers and customers, AI-based pest and disease diagnosis, decision support for planting and harvesting and knowledge-sharing collaboration dashboards

3.2.2 System Design

This design process included the creation of:Use Case Diagrams for the definition of user interactions, data Flow Diagrams (DFDs) to show the movement of data, engineered architecture diagrams and entity Relationship Diagrams (ERDs) for representing the structure of the database.

The system architecture diagram indicates interaction between components. Both logical and physical designs were developed to describe how system components communicate and operate.

3.2.3 Implementation (Coding)

This where the actual code of the system is done. Such system modules include AI diagnosis, market linkage, decision making, and dashboards which were then coded using specific languages or frameworks. Each module was designed as an independent unit before being integrated into an entire system.

3.2.4 Testing

After implementation, the functioning of each module in addition to the whole system was checked to ensure that everything works as intended. Types of tests conducted included: Unit Testing: Checking the components of the program individually, integration Testing: Ensuring that modules work well together, system Testing: Performance, reliability, and usability testing of this whole system, user Acceptance Testing (UAT): Satisfaction from farmers and agricultural experts regarding the system.

3.2.5 Deployment

On his state the system was hosted /deployed on Versel after configuration and setting done

3.2.6 Maintenance

The last stage involved continual monitoring and minor updating in order to correct errors, enhance performance, and implement user suggestion feedback. Maintenance assures the system's action, flexibility against emerging technologies, and security.

3.3 Tools and technologies used

To design and implement Agrilink AI intergrade system for farmers, The following tools and technologies will be needed:

3.3.1 Programming languages

JavaScript(Node.js,React)-for building the web interface and real-time dashboard

3.3.2 AI Learning Libraries

TensorFlow/PyTorch=for building and training AI model for pest and disease diagnosis

3.3.3 Database System

MySQL-for storing farmer profile, production data and diagnostic results

3.3.4 APIs and Framework

Restful APIs-to enable communication between modules

3.3.5 Frontend Tolls

HTML,CSS,JavaScript,React-for creating a responsive user interface

3.4 System Architecture

For real-time automated reasoning, streamlining memory, and seamless interaction between farmers, customers, and administrators, Agrilink's artificial intelligence consolidated structure is planned using a modular and scalable three-tier architecture. To enhance agricultural productivity and retail connectivity, it integrates machine learning systems, a data base, and synergies with a synergies purchasing display.

3.4.1 Overall System Architecture

The architecture consists of three main layers:

i. Presentation Layer

This layer provides a customer interface through which separate users (farmers, clients, and administrators) interact with the system.
Technologies Used: React.js, HTML, CSS, JavaScript, Bootstrap.
Functions: Farmer Portal: Crop image upload, viewing diagnosis reports., market Dashboard: Product listing and buyer access, decision Support Interface: Displays insights, weather data, and smart tips and collaboration Forum: Enables user discussions and information sharing.

ii. Application Layer

This will be the second brain of the framework, managing enterprise logic, inferring, and exchanging information between the front and back ends.

Technologies Used: Node.js, Express.js, TensorFlow, Python Flask (for AI microservice).
Functions: AI Diagnostic Engine – Detects crop diseases or pest damage using image recognition, market Linkage Service – Matches farmers to customers and markets, decision Support Engine – Provides intelligent farming recommendations and notification & Communication Service – Manages user alerts and messages.

iii. Database Layer

Stores all structured and unstructured data, including user details, crop images, and market transactions.
Technology Used: MySQL (Relational DBMS), Firebase (for images), Cloud Storage.
Data Stored: Farmer & Customer profiles, crop images and diagnosis results, market transactions and analytics and reports

3.5 Conceptual System Architecture Diagram

Table 1

Presentation layer
Farmer Portal ,Customer Portal ,Admin UI ,Market Dashboard ,Collaboration Forum

Table 2

Application Layer
Node.js Backend, AI Diagnostic Engine,Maket Services, Decision Support Engine, API Gateway

Table 3

Database Layer
MySQL Server, Cloud Storage, Backup DB

3.5.1 Use Case Diagram

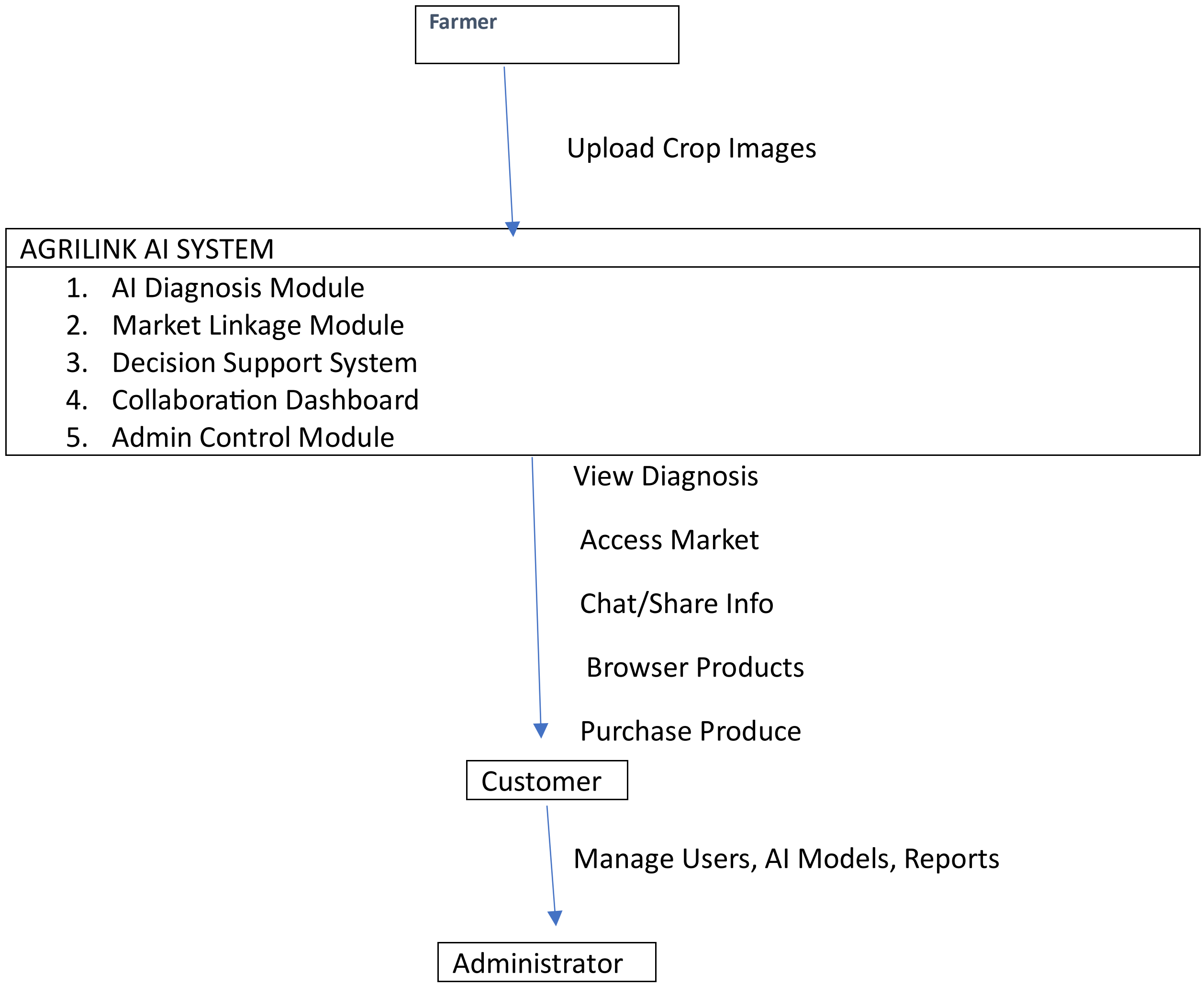

Figure 1

3.5.2 Data Flow Diagram (DFD)Level 0

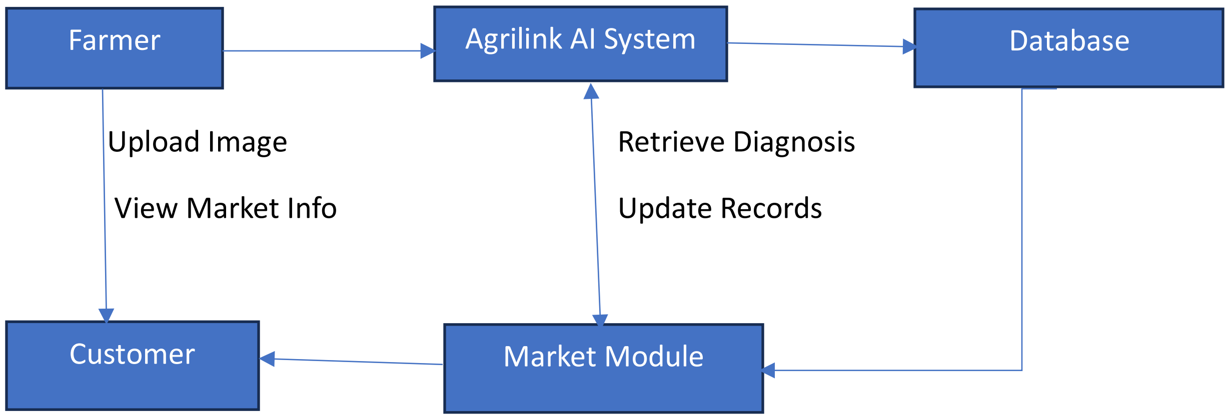

Figure 2

Main Flow:

Farmer upload crop image to the system, the AI engine processes the data and stores results in the database, market module retrieves data for buyers and decision support and collaboration module use stored data insights

3.5.3 Data Flow Diagram (DFD) Level 1 (AI Diagnosis Module)

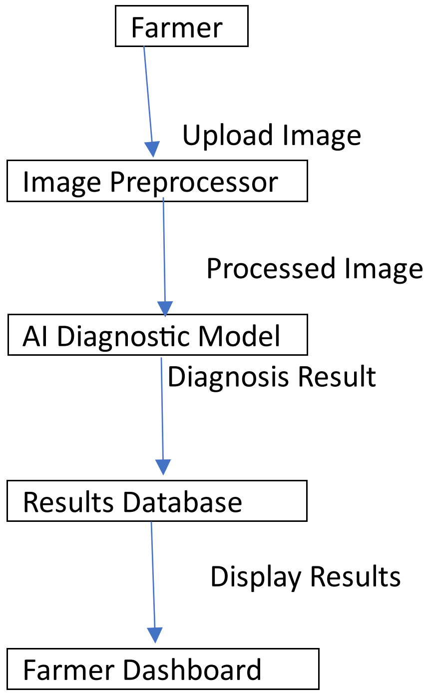

Figure 3

1.5.4 Entity Relationship Diagram (ERD)

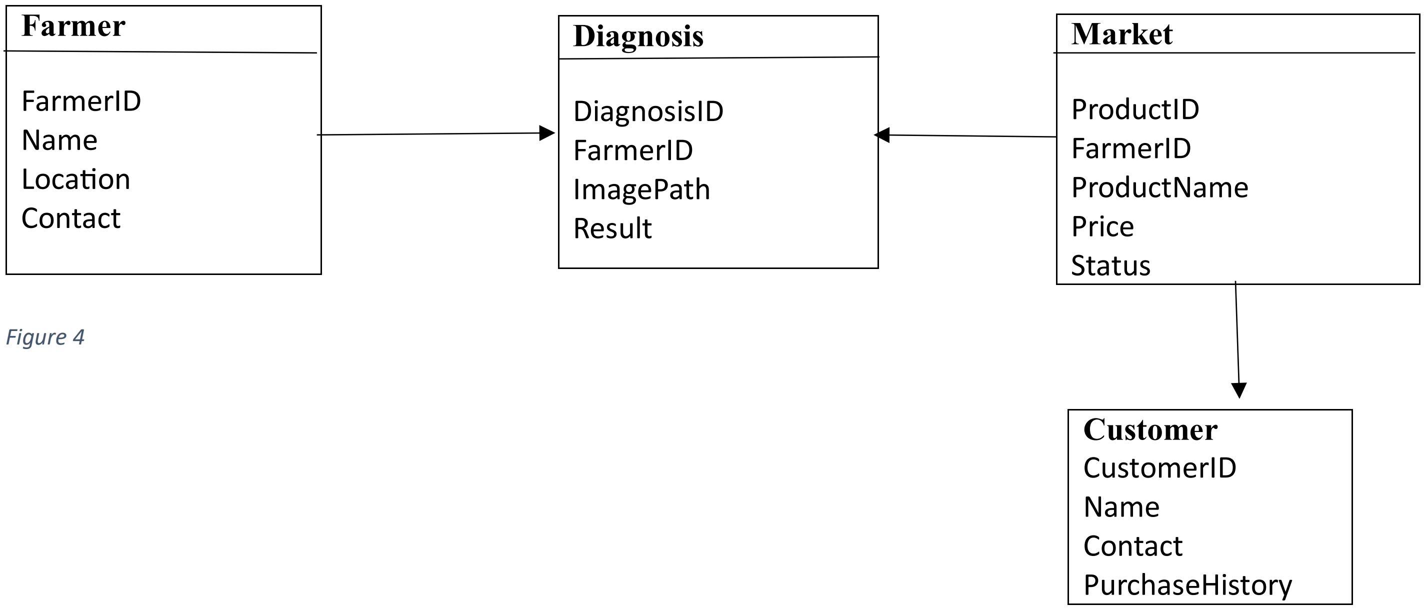

Figure 4

Relationship:

Farmer can have multiple Diagnoses and products, customer can purchase multiple products and diagnoses are linked to farmers through their IDs

Table 4

3.5.5 Logical Design Diagram

It show and represent how the system software components interact logically without showing the physical hardware

Table 5

Agrilink AI System
Modules: • User Management • AI Diagnosis • Market Linkage • Decision Support
External Entities:Farmers,Customers,Administrator
Databases: MySQL(User,Market,AI Results)

3.5.6 Physical Design Diagram (Network Architecture)

This shows how system components are deployed physically on hardware and cloud server

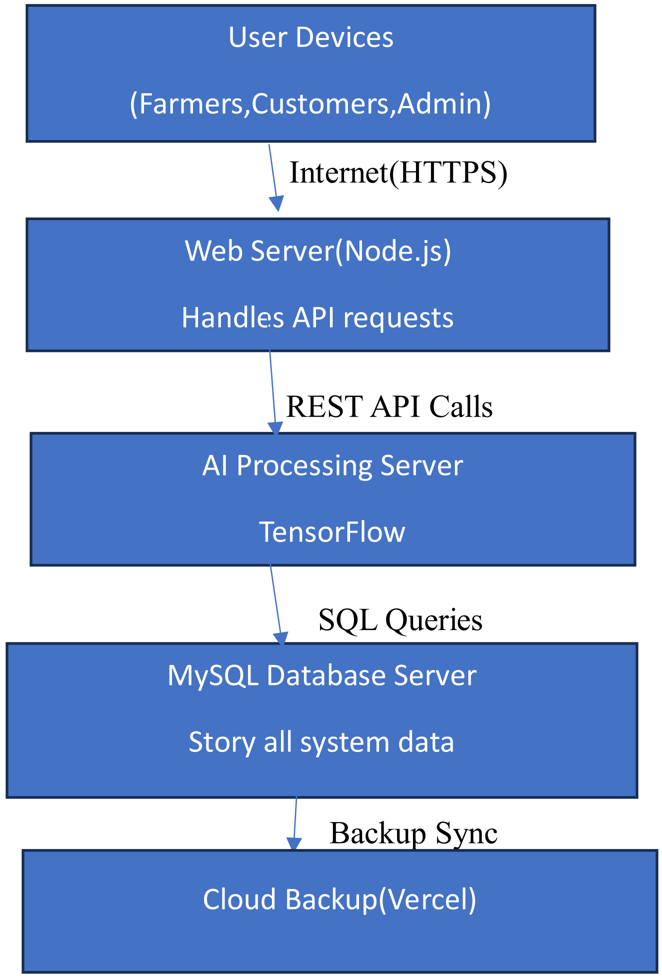

Figure 5

Explanation

Client Devices: Access the system via browser or mobile, web server: Hosts the user interface and APIs,AI Server: Runs pest and disease recognition, database Server: Store data and cloud Backup: Ensures data safety and recovery

3.5 Data Collection Methods

To ensure that the Agrilink Ai system will efficient and friendly to farmers ,both primary and secondary collection methods for data collection were used during research and development phases

3.5.1 Primary Data Collection

Primary information was obtained directly from the target users, i.e. Farmers, agriculturists, and retailers. The main approaches usually used are.

i. **Interviews**

Interviews were conducted with community farmers to understand problems with access to stores, pest control, and farmer supervision.
Example of Questions ask: What type of crop do you grow?, How do you currently identify crop diseases? And Would you be interested in using AI system for disease diagnosis?

ii. **Questionnaires**

The questionnaire was distributed to a sample group of farmers in order to gather structure information on their Digital Competences, Farming Habits, and preferred features in the agriculturist system.
Example of Questionnaire :What factors influence your buying decision?, Would you prefer a verified system that connects you to farmers directly? And Additional comment or suggestion

iii. **Observation**

In order to understand real-time obstacles such as pest outbreaks, weather effects, and postharvest management, monitoring of pasture activities was used.
The findings have led to the design of the Agrilink automated reasoning system, as well as the design of the user interface.

3.5.2 Secondary Data Collection

Secondary data was obtained from credible sources such as:

Academic journals, publications, and previous studies on AI in agriculture. Report on agricultural digitalization by companies such as FAO and World Depository. Online datasets for AI model training (e.g., open-source image datasets for plant disease detection).Documentation from existing AI agricultural platforms (e.g., Plantix, IBM Watson Decision Platform for Agriculture).

3.6 Data Analysis

Qualitative and quantitative analysis of the data collected. Qualitative examinations aimed at determining the characteristics of the subjects and the needs of the consumers, while quantitative examinations carried out as part of the measures examined the response and the organization's ability to provide services and accuracy.

3.7 Ethical Consideration in Data Collection

The purpose of the investigation into the compilation of earlier information was known to the participants. Confidential correction of all private and agricultural facts. Sharing was voluntary, and statistics were familiar with them purely for educational and development purposes.

3.8 System Implementation

The implementation consisted of integrating multiple system components into a working prototype. The core modules designed involve: AI diagnostics module- For the detection of crop diseases from uploaded images, market Linkage Module- This module links farmers directly with buyers. ,decision Support Module- These provides data-driven insight on farming activities And collaboration Dashboard- Here farmers can interactively share knowledge and experiences.

3.9 Testing and Evaluation

Testing ensured system reliability, functionality, and friendliness to the user, while evaluation focused on: Accuracy of the AI model on disease detection, responsiveness from the user interface, usability of the system and satisfaction of the farmers and market linkage and decision support efficiency.

3.10 Ethical Considerations

The study applied ethical considerations such that all data collected were strictly for academic and development purposes. The confidentiality of participants' information was maintained. All AI datasets utilized were open-source and were properly cited.

3.11 Summary

This chapter presented the methodology employed in developing the AgriLink AI system. The Waterfall model provided a structured and sequential method of development that guaranteed thorough documentation and quality during each of the stages.

References

(Zhang, 2021)

(Organisaion(FAO), 2022)

(Liakos, 2018) (Documentation, 2024)

(platform, 2023)u

Appendices

Appendixes 1:Sample Interview Question

1. What factors influence your buying decision?
2. Would you prefer a verified system that connects you to farmers directly?
3. Additional comment or suggestion
4. What type of crop do you grow?
5. How do you currently identify crop diseases?
6. Would you be interested in using AI system for disease diagnosis?

Appendixes 2:Sample questionnaire

Section A: Demographics

Name:

Location:

Type of Farming:[] Crop[]Livestock[]Mixed

Section B: Technology Use

Do you have access to the internet?[]Yes []No

Have you ever used a smartphone for farming purposes?[]Yes []No

If yes, which farming apps have you used?

Section C: System Expectations

Would you be interested in an AI system that diagnoses crop diseases? []Yes []No
Would you use a digital platform that connects you diagnosis crop diseases?[]Yes[]No
Wha features would you like to see in such a system?